Tic-Tac-Toe AI

A human-versus-AI game using Minimax search

PROJECT	TECHNOLOGY	AI METHOD
CodSoft AI Internship Task	HTML, CSS, JavaScript	Minimax + alpha-beta pruning

Project overview

For this project, I built a browser-based Tic-Tac-Toe game that lets a person play against an AI opponent. The player uses X and moves first; the AI uses O. I kept the project easy to run by putting the game in a single HTML file, with no installation required.

How the AI works

The AI uses the Minimax algorithm to look ahead at the possible moves and outcomes. It scores an AI win positively, a human win negatively, and a draw as zero. It then chooses the move with the best score. A depth adjustment encourages the AI to win sooner and to delay a loss.

Making the search more efficient

Alpha-beta pruning keeps track of the best options found so far and skips branches that cannot affect the final choice. This reduces unnecessary search while preserving Minimax's result. Because Tic-Tac-Toe has a small number of possible positions, the AI can search to the end of each game.

Features

The game reports wins, losses, and draws, highlights the winning squares, and keeps a score across rounds while the page remains open. The layout adapts to smaller screens and the board buttons include labels for accessibility.

What I learned

Building this project helped me understand how an AI can compare future game states and choose a move based on the outcome. I also learned how alpha-beta pruning makes a search more efficient, and how to connect game logic to an interactive browser interface.

How to run

Open **index.html** in a modern web browser. Click an empty square to play as X. Select **New game** to begin another round. No server or extra software is needed.

Project file: index.html | Supporting instructions: README.md